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Faculty of Engineering & Architectural Science

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Introduction

A small manufacturing company seeks to create a gearbox design that would attract hobbyists in the robot combat company. The company's rival has already created a gear box that is popular but not well liked by customers. The small company wants to use this as an opportunity to create a gearbox that is efficient and well structured so the transfer of the power from the electric motor to the robot's wheel is more functional in robotic matches. To create a better product, the gearbox created by the rival company can be reversed engineered to understand the problem with this design. This serves as a crucial step to increasing the small company's sales and profit. It is also crucial to consider the motor when considering the design for the gearbox which is an 8 mm shaft, 35.6 mm length, 2 mm keyway; #10-32 mounting pattern on a 2 in. bolt circle; 66 mm motor OD and maintain build quality under 350 oz·in stall torque.

The problem with the rival's existing gear box is that it is assembled inadequately, poorly transfers power and produced with a heavy material. The objective is to create a lighter, cheaper and fixed flaws gearbox. These changes will improve the gearbox design by decreasing power losses from the motor, making it easier for the battle bot to engage in combat and increasing the speed of the battle bot. It also minimizes costs which simplifies manufacturing. The constraints are that the gear ratio should remain the same, output shaft should be the same distance and orientation relative to the motor axis, output shaft should be compatible with the same robots and the function of the product should be the same. This will allow the battle bots to fight at its optimum speed and performance which will increase customer satisfaction.

The following sections are the background, problem analysis, design concepts, design selection and preliminary design which will improve the understanding of the problem. The background helps the reader understand the function of a gearbox and prior solutions used to conclude the final solution. The problem analysis breaks down the client's problem in detail. Design concepts explains the design alternatives of the gear box that were considered and explains the reasoning of why it was chosen. The design selection is the final design chosen from the alternatives with clear reasoning. Preliminary design lists the most important design parameters in the final design. Finally, the conclusion summarizes the important research, finds and final design solution.

Background

To efficiently compete in combat, it is necessary for robots to have a light and effective gearbox with functioning parts. This way the power from the motor of the robot will reach the wheels of the robot increasing its performance. The gear box is a transmissional mechanical device that transfers power from a rotating power source to another machinery, in this case the wheels. They ensure proper and optimal performance of different mechanical systems while handling different operating conditions [3].

The washers, bearings,shafts and gears in the housing of the gearbox reduce energy losses through friction, support misaligned shafts, change torque, speed and rotation, minimize vibration and shock load, which are important for a gearbox to function adequately [4]. Fixing the proportions, placements and changing the situated angles can further improve the performance of the overall gearbox.

The 20 degree pressure angle for the spur gear was chosen because it is the standard angle for all machinery. This specific angle provides great strength, balance and durability for the mechanical devices that were researched. The pillow bearing with a mounting system directly on the housing was decided due to reliability from previous designs. The pillow bearing with an interference/friction based mounting system was chosen due to the lower manufacturing costs mentioned from other designs while researching.

Problem analysis

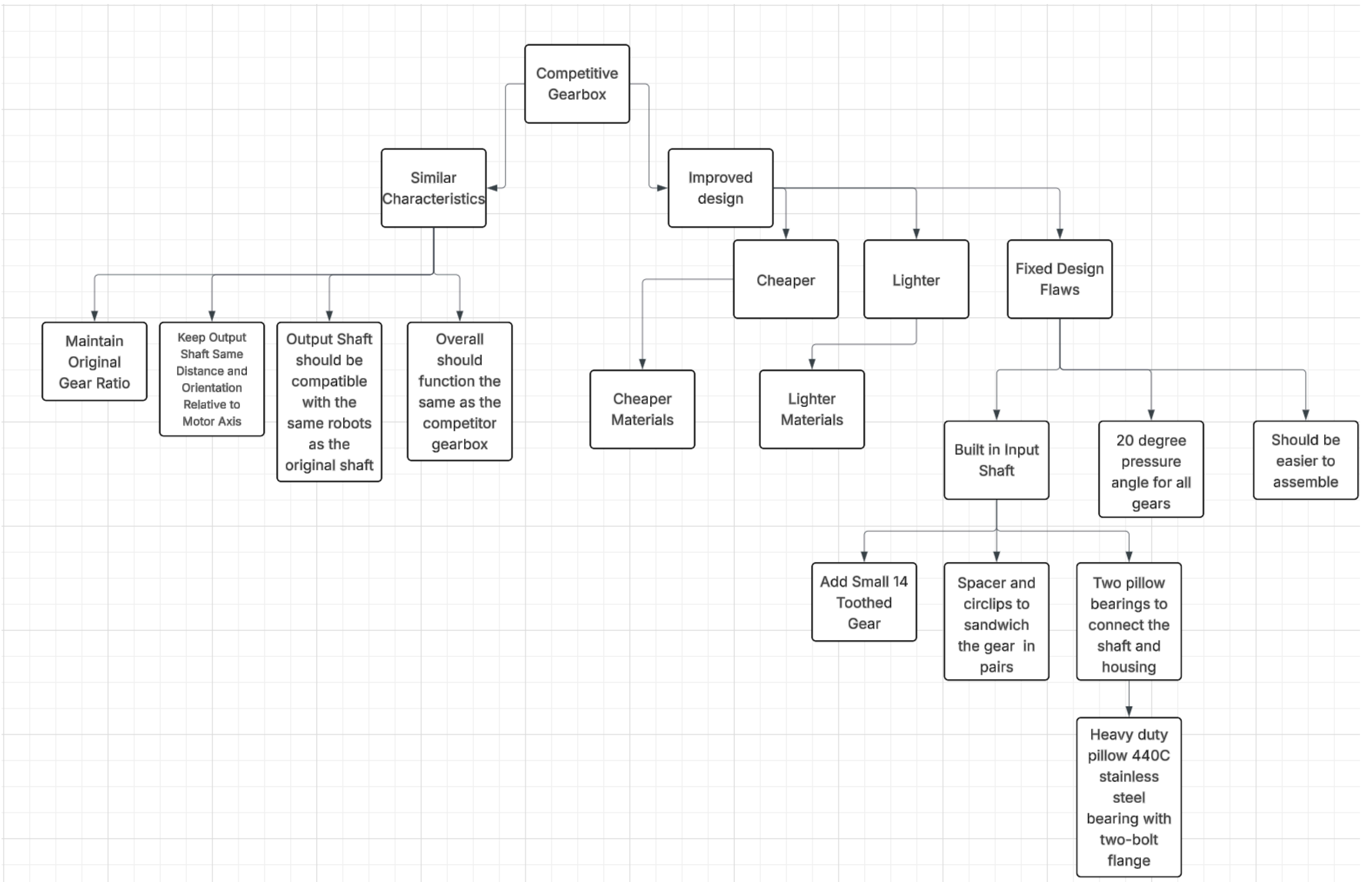


Fig 1. Flowchart via objectives, means, and constraints

Evaluation Metrics

Table 1. Similar Characteristics

Are all required similarities stated in the problem statement met?	
Yes	Satisfactory
No	Unsatisfactory

Table 2. Gearbox Cost

Percentage of competitors cost	
> 150%	Intolerable
> 100%	Not Ideal
= 100%	Acceptable
< 100%	Ideal

Table 3. Weight

Percentage of competitors weight	
> 125%	Intolerable
> 115%	Unsatisfactory
= 100%	Acceptable
< 100%	Ideal

Table 4. Ease of Assembly

Time required to assemble or disassemble the gearbox	
≥ 45min	Unsatisfactory
≥ 20min	Not Ideal
> 10min	Acceptable
≤ 10min	Ideal

Design concepts

Ideation

To develop various design concepts, a morphological chart can break down the various redesign ideas into categories. Because clients continuously critiqued the slop, aspects like alignment, mesh efficiency, and durability will be the focus of design considerations. Thus, the options below also assume that the geometric constraints in the project brief are met in the design. The result is a ready-to-use gearbox that can directly interface with existing battle bots.

Table 5. Morphological Chart Regarding Potential Design Concepts

Gearbox Category	Option A	Option B
Material for housing	6016-T6 aluminum for lower costs	7075-T6 aluminum for additional strength
Pressure angle	20° stock gears (module 1.27) mm maintained)	20° stock gears (module 1.27 mm maintained)
Input shaft, output shaft, and gear material	4140 steel (better degradation resistance) for input shaft and gear, keep output shaft the same for easy bot integration	A36 steel (focus on cost, abundance, and ease of manufacturing) for both shafts, keep 4140 steel gears
Bearing for input shaft	Pillow bearing with a two screw mounting system directly onto the housing for secure attachment	Pillow bearing with an interference/friction based mounting system into the housing walls for lower cost
Fastener methods	Use one screw type (simplify assembly and repairs)	Keep screws unchanged (works well, better reliability)
C-Ring	Redesign output shaft to keep gear in place without a part needing to be added on.	Design another separate piece that attaches and detaches easier onto the input shaft. It would be a new iteration of the C-ring that functions the same with a different installation method

Concept A Rational

There are two main conceptual designs. Concept A, through the series of options in column A, pivots around delivering a cheaper, easier-to-manufacture, and simpler gearbox that reduces sources of slop and efficiency through optimal bearing and gear mesh compatibility. This keeps an assembly that is easy to repair.

Concept A consists of 6016-T6 aluminum housing, a cost-driven choice with dimensions that pertain to the original, including hole sizes. This aluminum alloy is cheaper compared to alternatives and lighter, which facilitates a lower cost. Although the drawback is lower stiffness, the yield and tensile strengths (210MPA and 280MPA, respectively) are more than enough to discourage housing flex. However, the corrosion resistance, weldability, and machinability score also provide the benefit of cheaper and easier manufacturing than the original material.

Currently, the gears are manufactured from 4140 steel, while the output shaft is made of aluminum 2024. The choice of 4140 steel for gears is consistent with material specifications, which emphasize good wear resistance (tensile strength between 655-1230 MPa, yield strength between 415-925 MPa, and excellent machinability). The aluminum 2024 output shaft will remain unchanged, given the nominal strength-to-weight ratio. It also ensures easy integration with existing battle bots. The input shaft, under much of the same stress as the gears, will also be made from 4140 steel, which aligns with steel properties. Steel has a better load capacity than aluminum, which is key given the cylindrical-to-square-end input shaft constraint. Moreover, a direct connection to the motor corresponds to stress from torque, which makes this choice beneficial. These changes reduce the potential for deformation, slop, and inefficiencies.

Regarding the gears, the module (1.27mm) and gear ratio will remain unchanged. However, all pressure angles will be 20°. The current pressure angle of 14.5° leads to severe undercutting, poor shock resistance, and low wear resistance, which contributes to degradation and a "sloppy" assembly. This is also exaggerated due to the low tooth count. Under a 20.0° pressure angle, gears will have thicker tooth roots, resulting in more resistance to wear and tear. This stems from less undercutting and a higher tolerance for torque. This significantly reduces slop by better sustaining the mesh between gears, vital when a few mm makes a huge difference. Despite the larger loads on the bearings, the change is worthwhile, as bearings can be redesigned. There is also the bonus of abundance: many gears and machinery are available at 20.0° compared to 14.5°, which keeps the gearbox even more cost-effective by decreasing manufacturing costs.

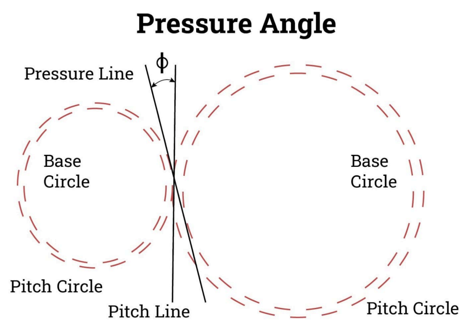


Fig 2. Illustration of pressure angles in spur gears [1]

As part of the redesign, the shaft geometry has already been outlined (25mm from the housing and a 6.5 mm × 6.5 mm square profile over 5mm of length). Thus, the main design choice comes down to mounting the shaft. An optimal choice is to use two pillow bearings that connect to the shaft and housing on either side. The design constraint of a 0.375in shaft diameter narrows bearing options to a heavy-duty pillow bearing with a two-bolt flange mounting system from McMaster-Carr [5]. The bearing design uses 440C stainless steel, excellent for intense and prolonged loads, which also addresses the additional load on the bearings due to the pressure angle change from earlier. The seal ensures consistent performance over time, while the bearing itself has a mounting system that uses two bolts to attach directly to the face of the gearbox. In conjunction with the heavy-duty press-fit mount between the inner ring and shaft, these changes ensure further precise radial alignment, which significantly addresses the inefficiencies and slop. This is important when a few mm of misalignment is critical.

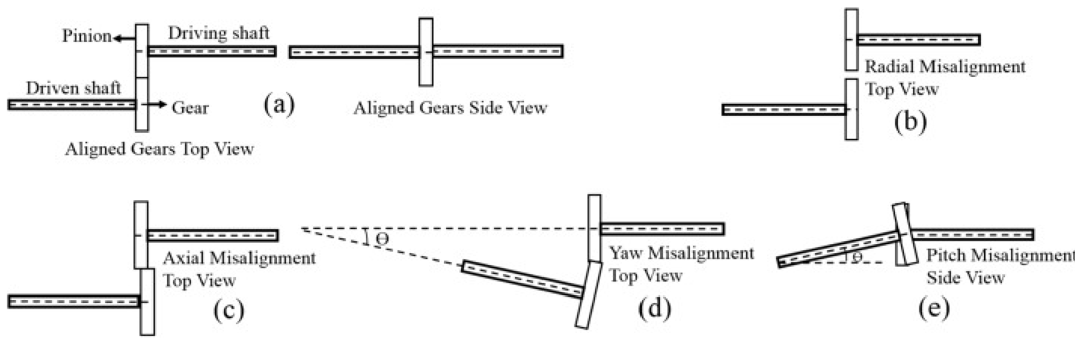


Fig 3. Visual showcase of gearbox misalignment [2]

Regarding the assembly, standardized screws will fast-track assembly and repairs. This not only further reduces costs but also ensures assembly is consistent throughout, which reduces the possibility of misalignments and errors in manufacturing. The input shaft will have a spacer, and circlips sandwich the gear in pairs, with both on either side, much like the existing setups for the current gears. As a result, the input gear will preserve axial alignment with the mating gear, which improves mesh. This further reduces the remaining inefficiencies and slop during reversals and torque under operation. However, under concept A, the output shaft will be replaced with the same output shaft except instead of possessing an indentation to house the C-ring a ridge will exist there instead. This will keep the side gear where it needs to be, removing the necessity of the C-ring which was difficult to install and broke easily. This design change also requires changing the small side bearing, giving it a bigger upper lip and screw holes so it can be attached from the outside. This should have little to no effect on the function of the output shaft and will greatly improve assembly and disassembly experience, saving time and the need for replacement C-ring. Images for side pin and small side bearing are below

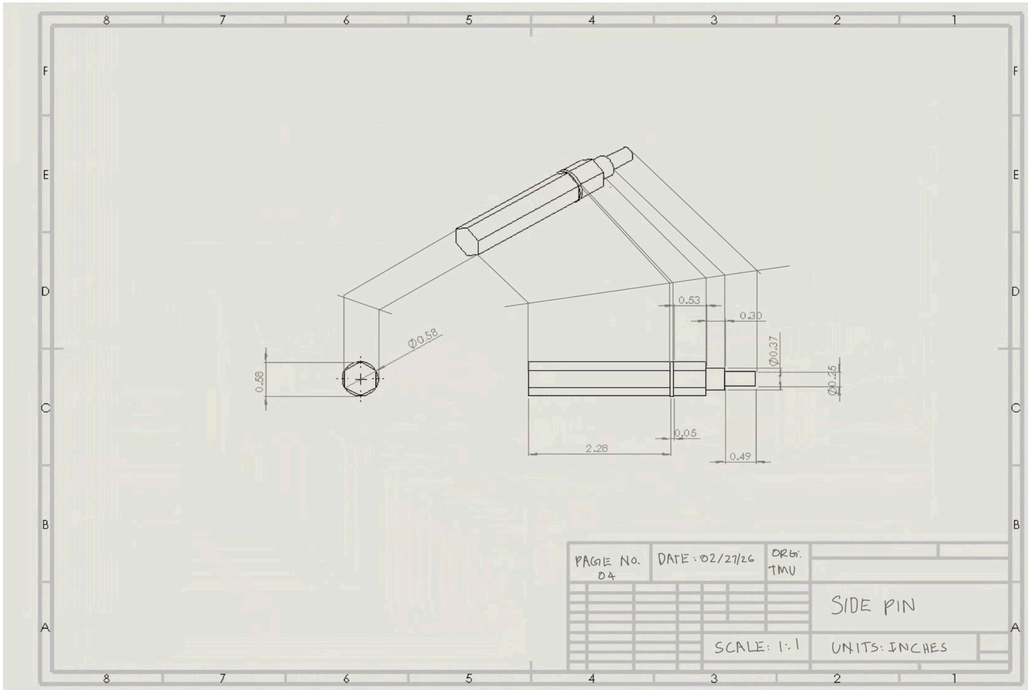


Figure 4. Side pin

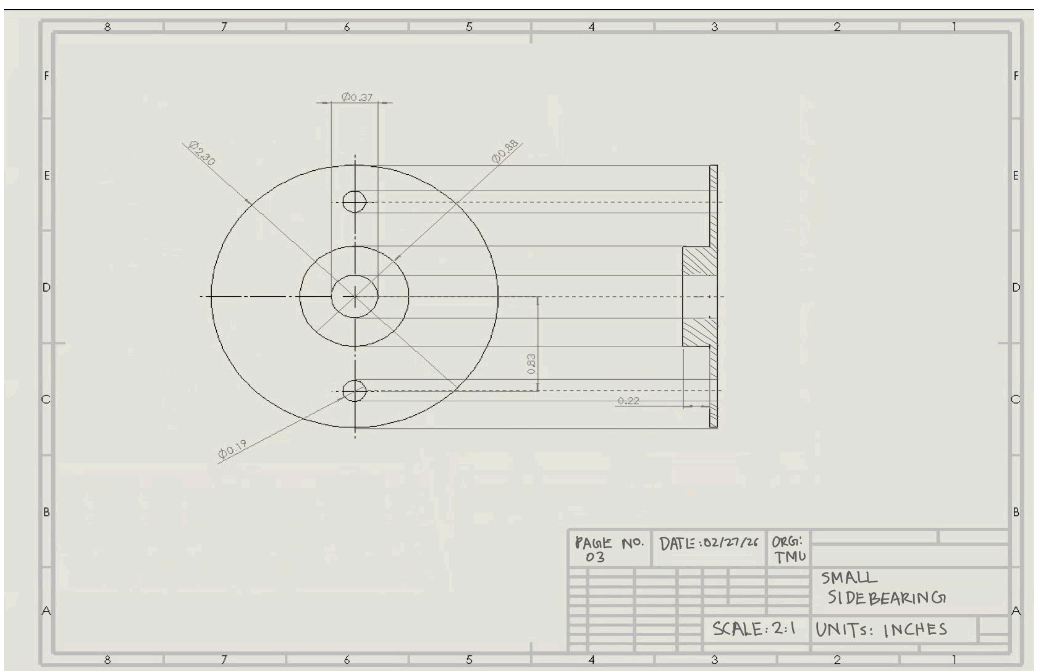


Figure 5. Small side pillow bearing

Concept B Rational

Concept B focuses on an approach that revolves around reducing modifications while investing in a few aspects, most noticeably the materials. This makes redesign simple, highly doable, and integrable with current battle bots with absolute certainty.

In this configuration, option B uses 7075-T6 aluminum to minimize housing flex. This particular aluminum has a tensile and yield strength of around 500MPA, which is far more than capable of handling intense loads and the additional forces applied via bearings, bores, and screws. In doing so, it ensures hyperaccurate alignment through minimal deformation along regions under heavy loads. The result is minimal repairs required by the client, which leads to a cheaper and more reliable product, particularly for the user. However, this aluminum is incredibly expensive and unmachinable, which makes manufacturing costs more expensive than the costs saved. This is a major drawback, as the project brief specifically asks for a price-competitive product. A price increase is not a drawback that the product can take.

4140 steel for the gears is a baseline change for both concepts A and B. On the other hand, A36 steel for input and output shafts can help offset the cost and machinability issues, as A36 steel is much cheaper and more machinable compared to steel and aluminum counterparts. However, the resultant is very high malleability under shock loads and thus, shafts can bend and twist under intense pressure. This is a drawback that can not be absorbed, given the nature of the project and the need for better alignment.

Maintaining the 20° pressure angle is advantageous and serves as a baseline change for both concepts A and B.

Keeping the fasteners the same as the original simplifies the redesign process, which simplifies manufacturing, as the process has already been proven capable, as seen by the current gearbox company. This also ensures that the gearbox will perform. However, our product must be competitive, and the earlier swaps in material, stress yield, cost, manufacturing, and quality issues would make it uncompetitive compared to the existing design. Regarding the C-ring, it is a very inconvenient piece of the gearbox that is difficult to put on, remove, and is prone to deformation. The best way to fix this while having minimal design changes to the output shaft section is to come up with an alternative C-ring design. One possibility is to have more of a clamp that has some sort of locking mechanism like a bracelet or earring would.

Design selection

Decision Matrix

To decide between the potential design options, the first step is to develop a weighted decision matrix to accurately score both design options.

Table 6. Weighted Decision Matrix for Concept A and B

Metric	Weight	Concept A Score	Concept A Weighted	Concept B Score	Concept B Weighted
Cost	0.3	4	1.2	2	0.60
Assembly → slop reduction	0.25	4	1.0	3	0.75
Efficiency → mesh quality	0.20	4	0.80	3	0.60
Durability under load	0.15	4	0.60	3	0.45
Manufacterability and reparability	0.10	4	0.40	2	0.20
Total	1		4.00		2.60

Decision Rationals

Regarding cost, aluminum 6016-T6 in concept A is more cost-effective than aluminum 7075-T6, which is more expensive due to material properties. On the other hand, the use of A36 steel for shaft design in concept B is more cost-effective than 4140 steel in concept A. Moreover, it is far more machinable due to the malleability, which drives down manufacturing costs. This would give concept B a higher score if it were not for the friction-based mounting system for the bearings, which increases costs via additional tolerance control. Thus, the overall price of concept B is far greater than that of concept A, hence concept A scores higher.

The component that yields the largest influence over assembly precision is the bearings. Both concepts use the same pillow bearings but different mounting systems. Concept A uses a two-bolt flange mounting system, which is acceptable given the shocks, vibrations, and torques. It securely mounts the bearing to the housing. Concept B uses an interference fit, relying on friction between the bearing and walls of the housing. This method is particularly sensitive to tolerances in manufacturing and can deform the bearings, which ruins alignment under loads. Thus, although a friction-based mounting is generally acceptable, given the context of the design, it is not optimal. Consequently, option A scores considerably higher. Concept A is also necessary for the concept A for C-ring changes.

Examining efficiency depends on the mesh between the gears. The mesh is highly altered depending on the pressure angle. As previously mentioned, 20° pressure angles serve as a baseline improvement. However, the bearing mounting method for concept A ensures that the mesh between the gears is better than that in concept B; thus, concept A has a higher score.

Durability is determined by the materials. Concept A uses 4140 steel for the input shaft and gears, which provides the ability to withstand power transmission. The housing consists of a less durable, but more than adequate 6016-T6 aluminum, which will allow for minor but insignificant housing flex. Overall, durability is acceptable. Aluminum 7075-T6 is far more durable and ensures no housing flex in any region under load. However, A36 steel for the shafts is malleable and will warp considerably under intense shock loads. Thus, concept A has the benefit of being a generally more durable design across the board, compared to concept B's extreme durability for the housing but very poor durability for the shafts. Consequently, concept A has a higher score.

Manufacturability and reparability ultimately come down to fasteners. Concept A standardizes these under a single type of screwtype, while Concept B maintains the current sets. Moreover, the retention method in concept A disregards the C-ring in the output shaft, unlike concept B. Previous experiences disassembling the product made it apparent that the C-ring was a hassle to repair, undo, and warp under pressure easily. In conjunction with benefits from using a single fastener, it is clear that concept A is more serviceable and manufacturable.

The total scores tally up to 4 compared to 2.6 for concepts A and B. Thus, concept A is clearly the better option and is the one that is chosen. The weighting for each section comes down to what the clients asked for, which was a more efficient and less sloppy gearbox at a competitive price. The result is that cost, assembly, and efficiency are weighed the heaviest, while important complementary aspects like manufacturability and durability are weighed less.

Preliminary Design

Our preliminary design consists of these changes:

- Housing material was changed to 6016-T6 aluminum. This sacrifices an acceptable amount of durability for a decrease in costs.
- Input shaft and gear materials were changed to 4140 steel for increased durability.
- Altered gearbox screw holes so only one type of screw is needed. Makes assembly and replacement easier.
- Gear pressure angle was changed to 20^{o} from 14.5^{o}. This puts less strain on the gears. This change is not shown in the CAD.
- The C-ring was disposed of by altering the output shaft and small side bearing so the output shaft could have a permanent ring in place of the C-ring. Replacing the fragile C-ring saves a lot assembly and disassembly time and is one less part that needs replacing.

CADs of final design are below.

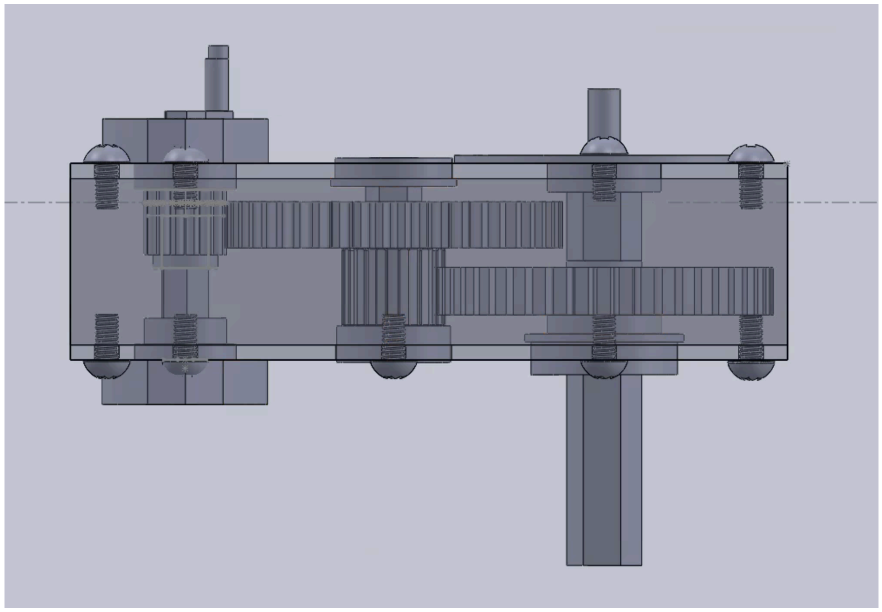


Figure 6. Final CAD side view

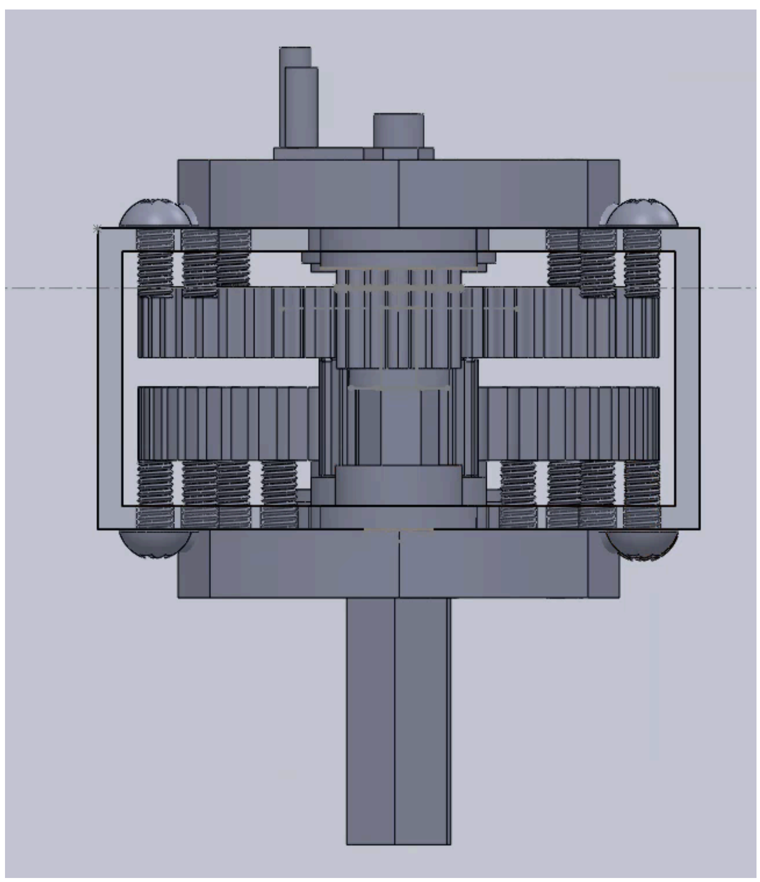


Figure 7. Final CAD front view

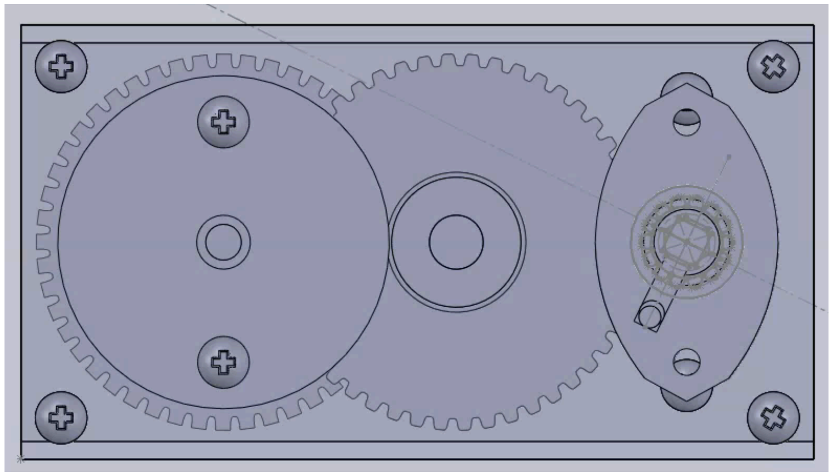


Figure 8. Final CAD top view

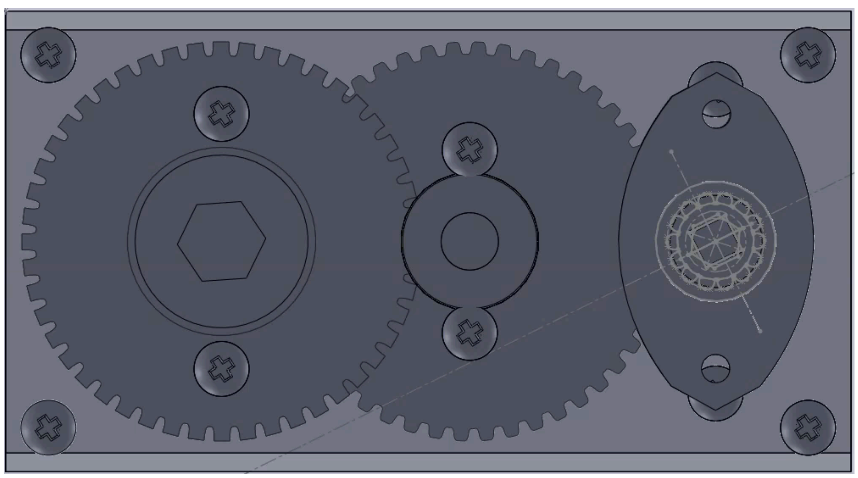
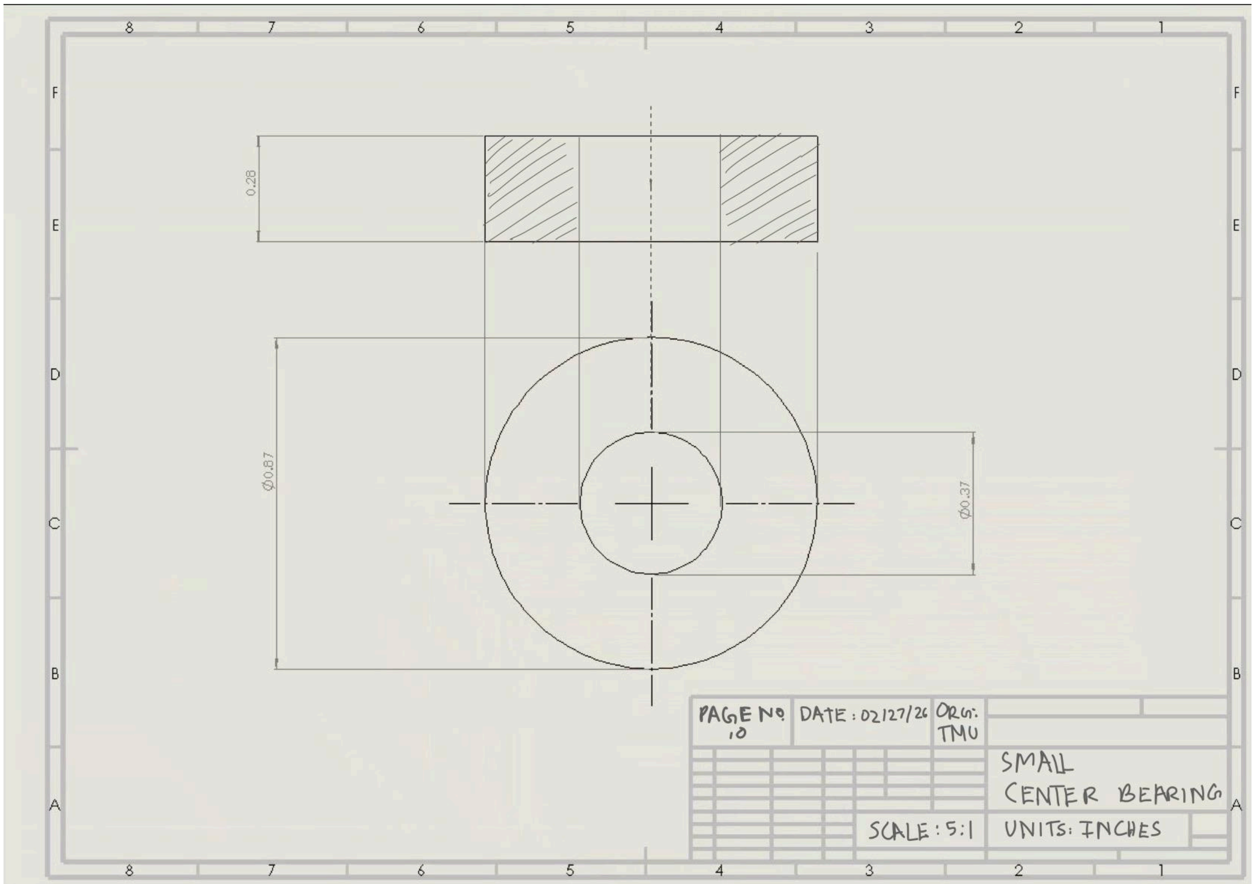


Figure 9. Final CAD bottom view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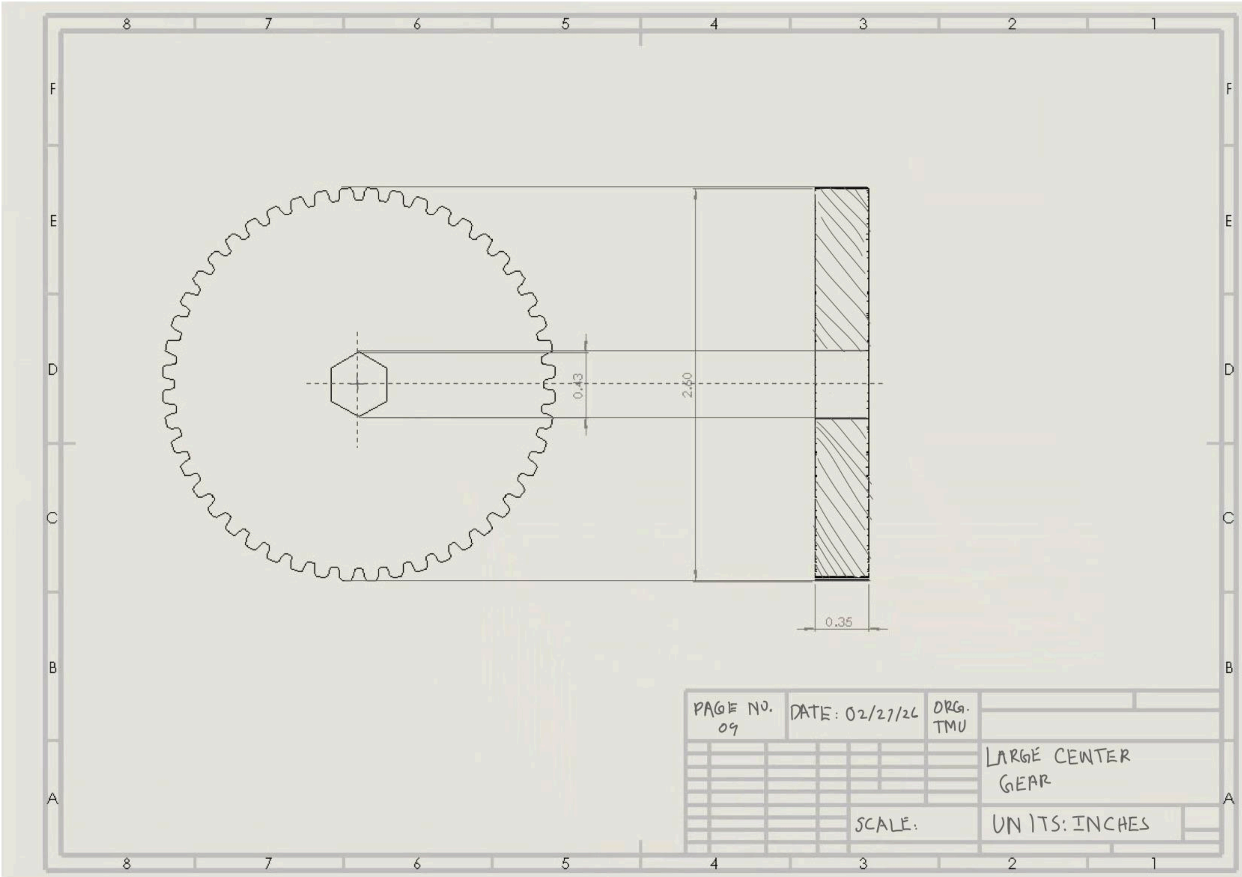
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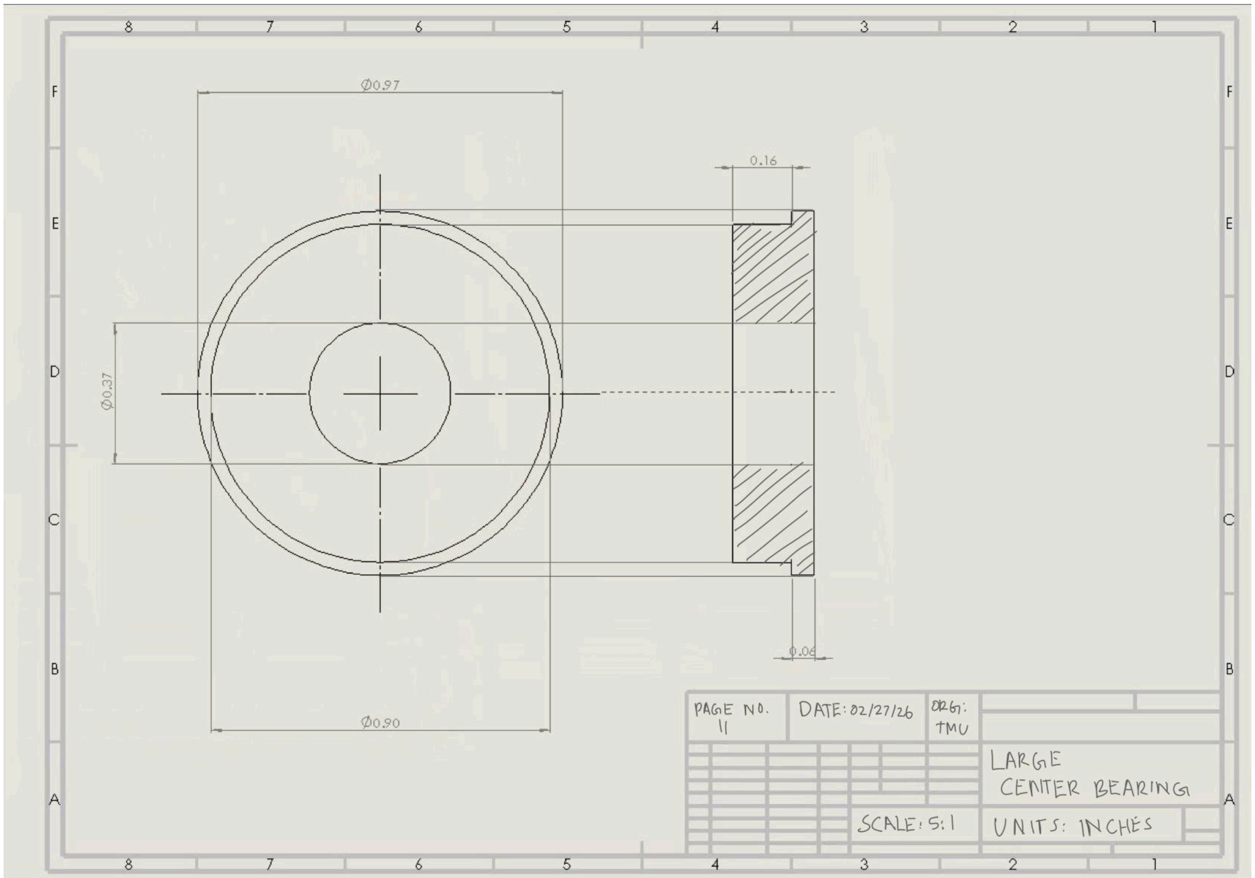
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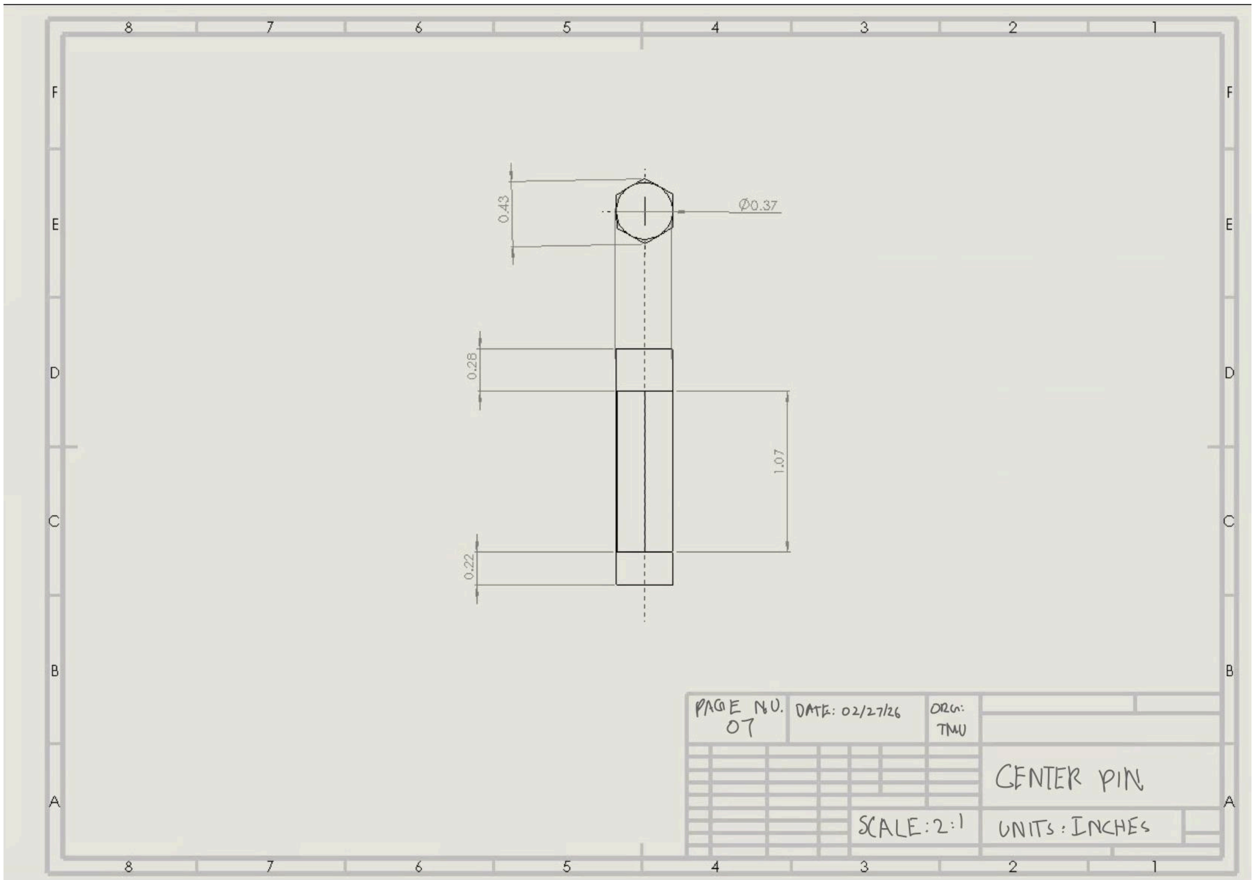

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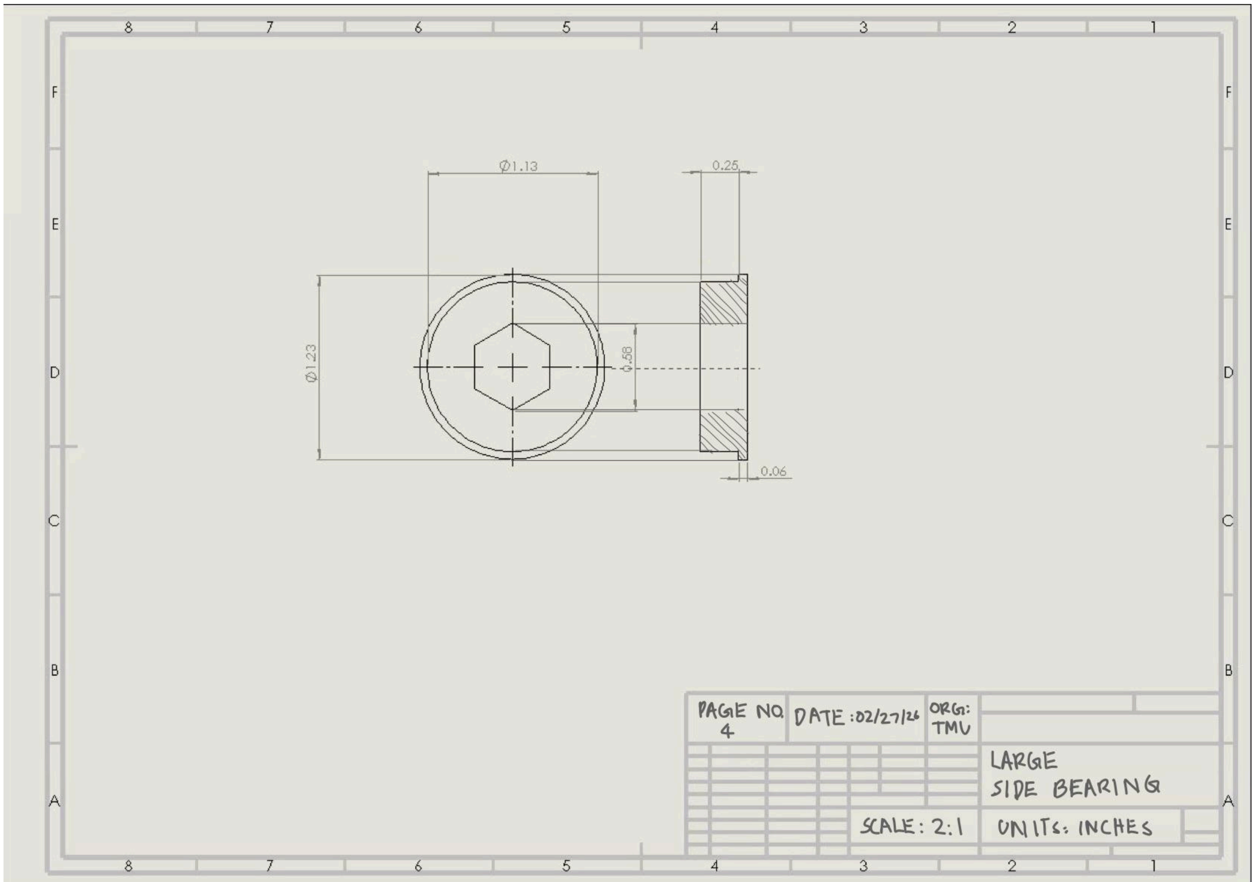
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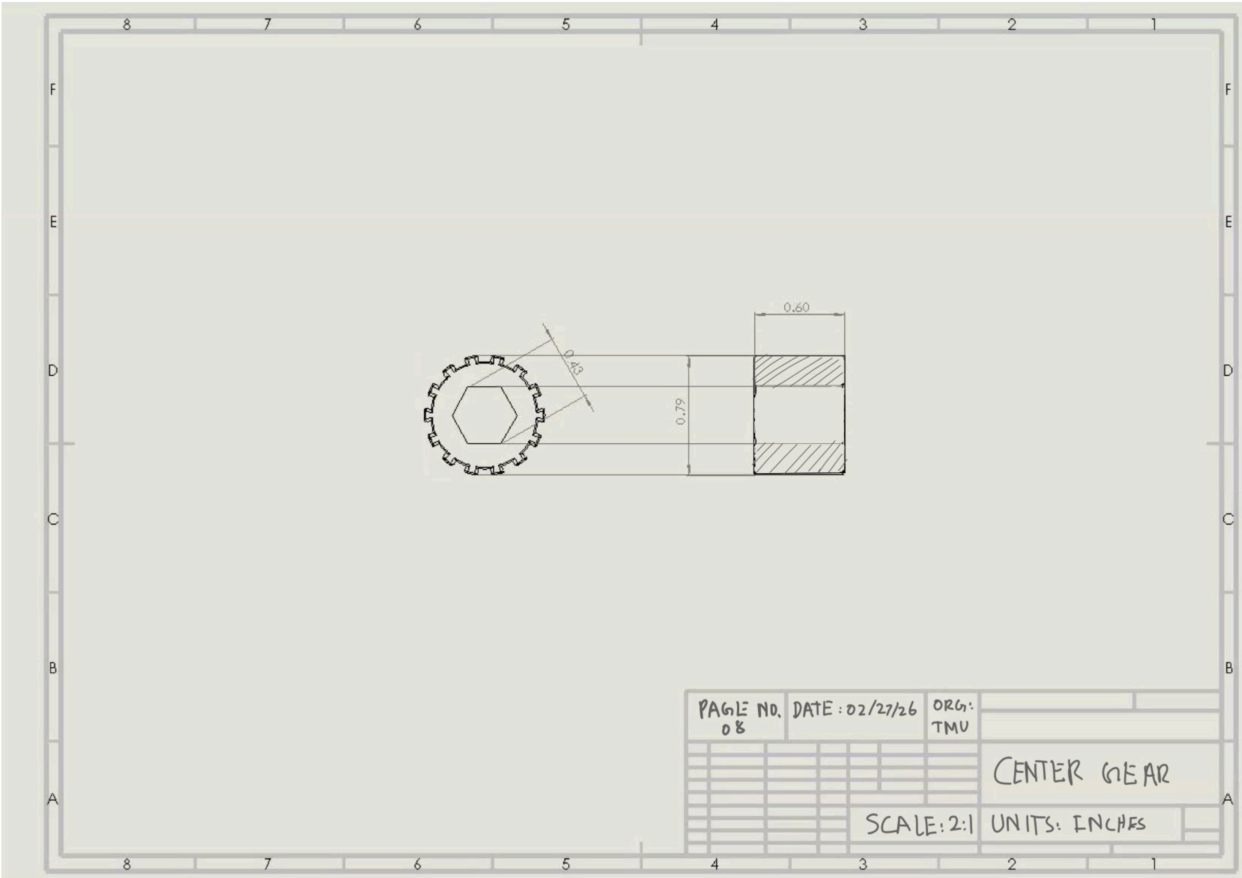
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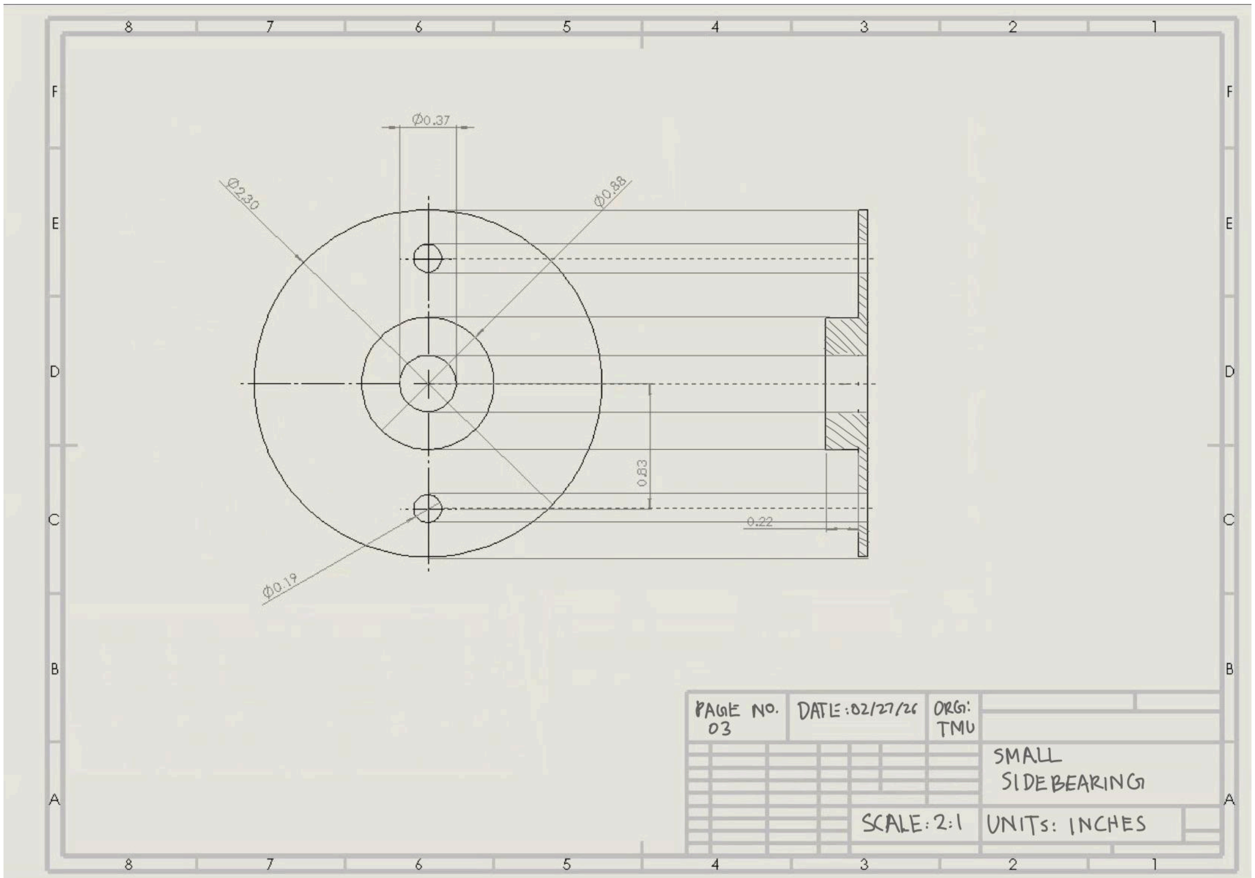
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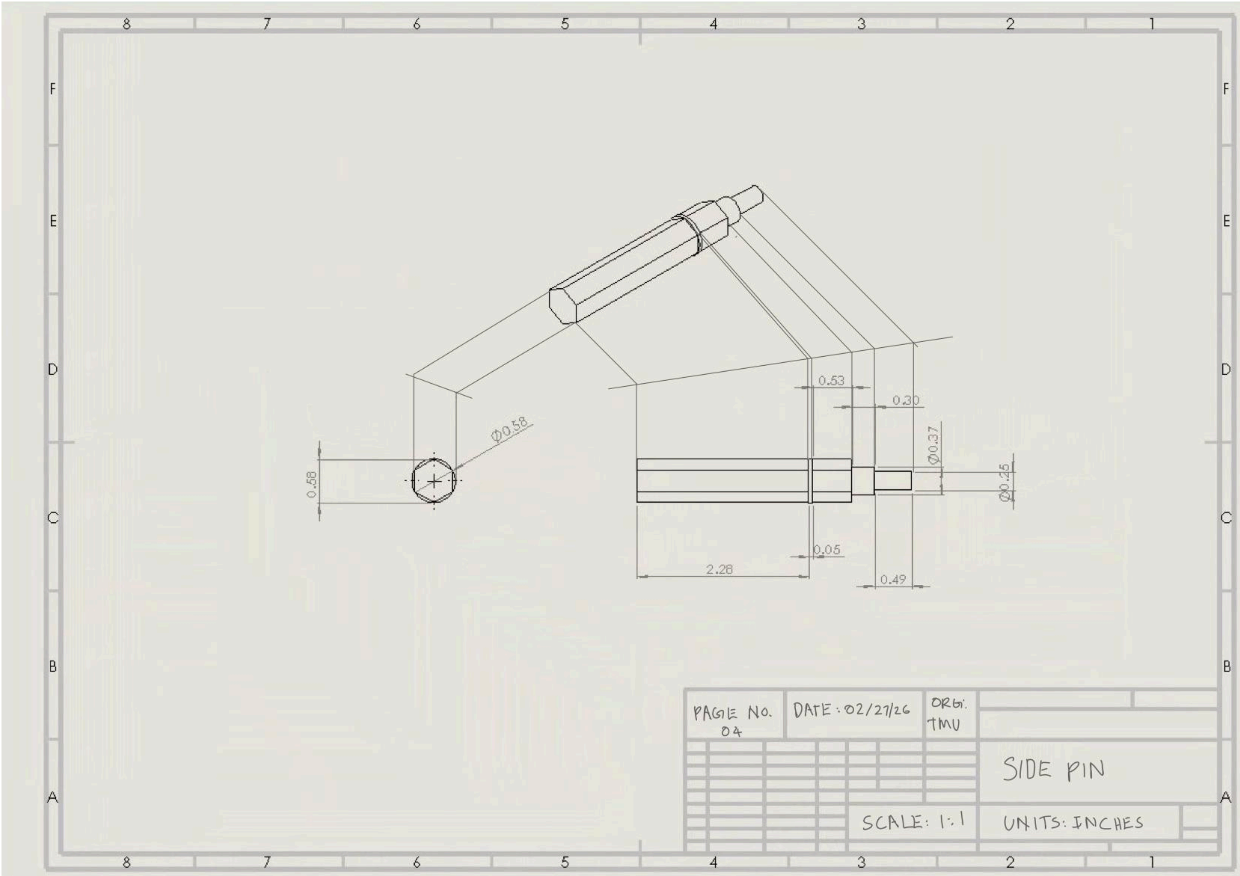

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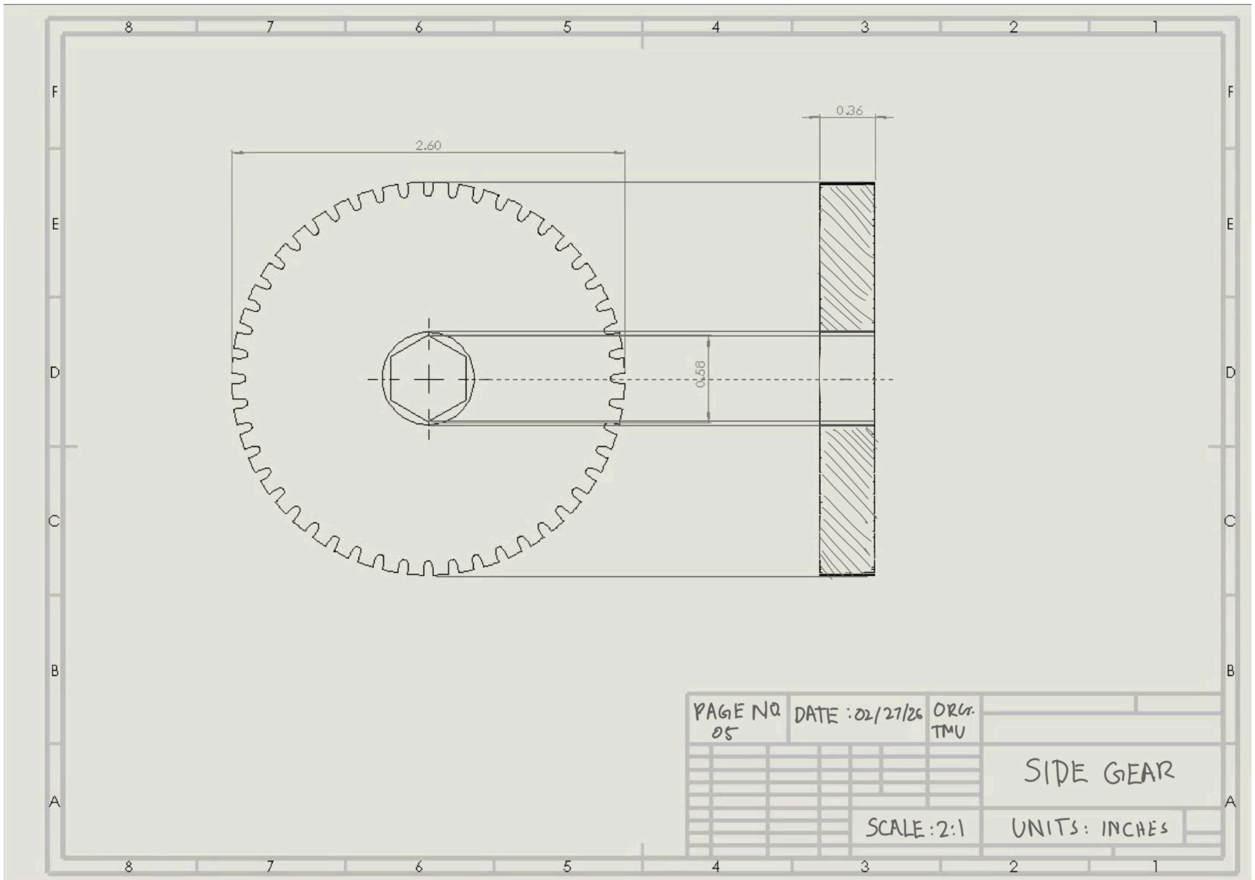

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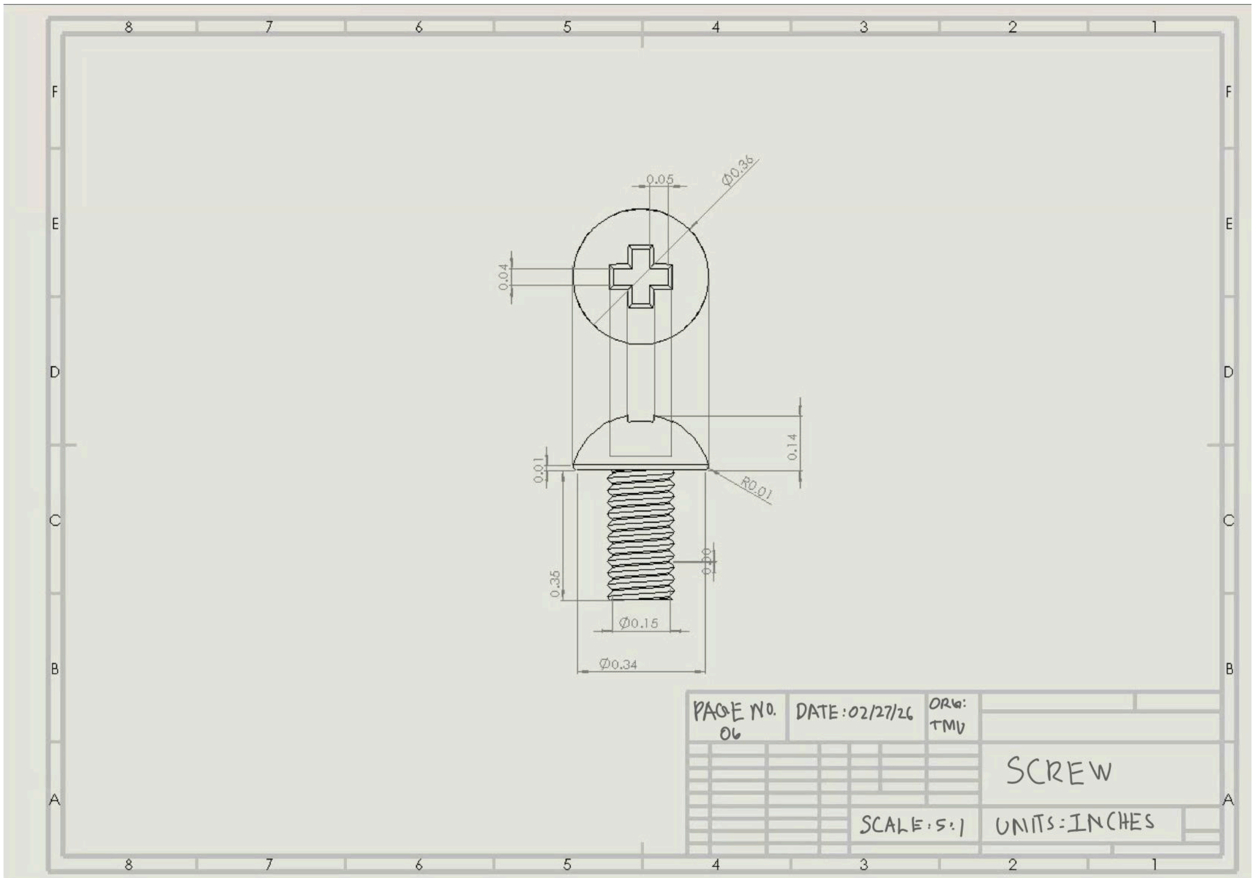

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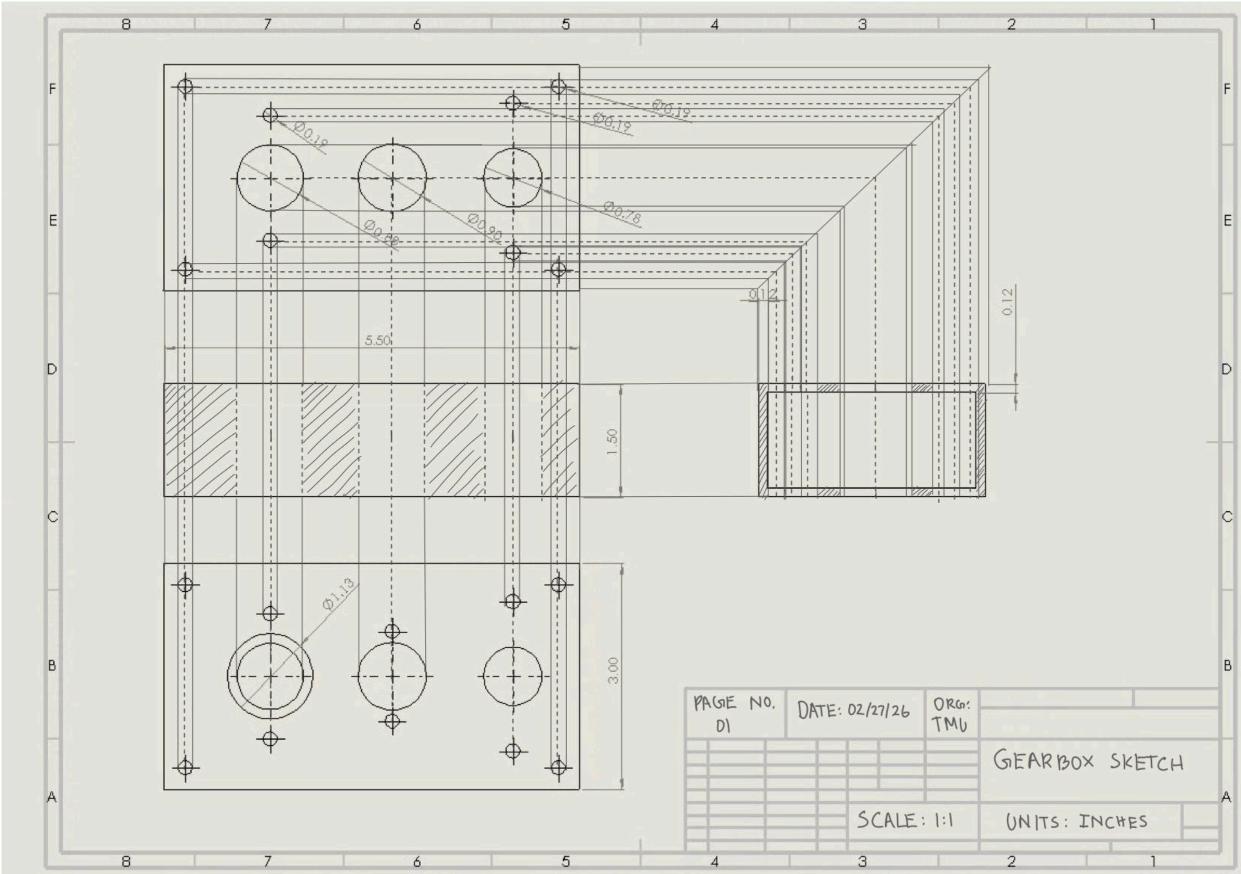
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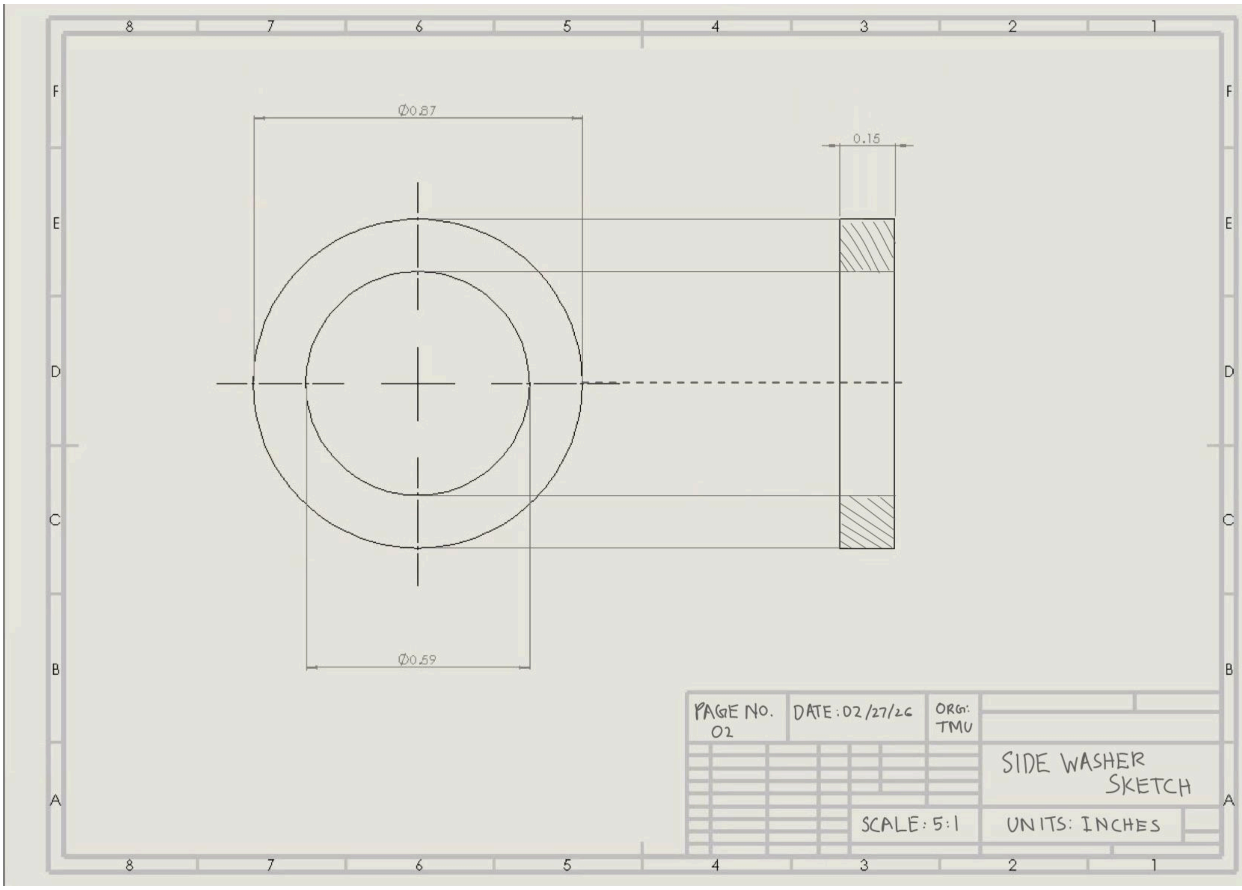

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Conclusion

This project redesigned a battle bot gearbox to produce a competitive product that can be integrated with battle bots on the market immediately. This new version resembles the original, with key modifications that improve cost, alignment, and efficiency. Examples of constraints that had to be met during redesign includes the following: maintaining the gear ratio, shaft interface, motor mounting geometry, operation capabilities of 350 oz in stall torque, a new input shaft that protrudes from the opposite side of the output shaft (25mm from the housing), and a 14-tooth input gear.

Brainstorming took the form of a morphological chart. With a set of design options, a weighted decision matrix was created to finalize a design option for the preliminary design. The selected concept was option A, given its low costs, strict alignment, manufacturability, and durability. The crux of these benefits stems from a cost-effective and performative housing material (aluminum 6016-T6), 20° pressure angles for nominal mesh, 4140 steel for gear and input shaft material to prevent degradation, and mounted pillow bearings to support the input shaft with strong alignment. The addition of a redesigned circlip system alongside standardized screws helped further manufacturability.

For the preliminary design, a final CAD model, with an assembly, orthographic drawings, and geometric constraints, was created. This model helped showcase the specifics and how the final design would look when retrofitted with the new modifications.

There are various remaining steps for this project to be fully completed beyond the deliverables. One of the most notable steps would be to focus on finite analysis to evaluate the structural integrity of the design, especially given the numerous material changes that were made. This would ensure the final project could realistically operate under battlebot conditions.

Manufacturing would also be the next step, also known as prototyping. This would allow us to test the product to examine issues with manufacturing on a commercial scale that would otherwise go unforeseen. Overall, the project was a success, and the final product met the satisfaction of the brief.

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